

```

        return myMethod(num - 1) + num;
    }
}

```

6. Convert the following iterative method to one that uses recursion:

```

public static void sign(int n)
{
    while (n > 0)
    {
        System.out.println("No Parking");
        n--;
    }
}

```

7. Write an iterative version (using a loop instead of recursion) of the factorial method shown in this chapter.

Short Answer

1. What is the difference between an iterative algorithm and a recursive algorithm?
2. What is a recursive algorithm's base case? What is the recursive case?
3. What is the base case of each of the recursive methods listed in Algorithm Workbench 3, 4, and 5?
4. What type of recursive method do you think would be more difficult to debug: one that uses direct recursion or one that uses indirect recursion? Why?
5. Which repetition approach is less efficient: a loop or a recursive method? Why?
6. When recursion is used to solve a problem, why must the recursive method call itself to solve a smaller version of the original problem?
7. How is a problem usually reduced with a recursive method?

Programming Challenges

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1. Recursive Multiplication

Write a recursive function that accepts two arguments into the parameters *x* and *y*. The function should return the value of *x* times *y*. Remember, multiplication can be performed as repeated addition as follows:

$$7 * 4 = 4 + 4 + 4 + 4 + 4 + 4 + 4$$

2. isMember Method

Write a recursive boolean method named `isMember`. The method should search an array for a specified value, and return `true` if the value is found in the array, or `false` if the value is not found in the array. Demonstrate the method in a program.

3. String Reverser

Write a recursive method that accepts a string as its argument and prints the string in reverse order. Demonstrate the method in a program.

4. maxElement Method

Write a method named `maxElement`, which returns the largest value in an array that is passed as an argument. The method should use recursion to find the largest element. Demonstrate the method in a program.

5. Palindrome Detector

A palindrome is any word, phrase, or sentence that reads the same forward and backward. Here are some well-known palindromes:

Able was I, ere I saw Elba
A man, a plan, a canal, Panama
Desserts, I stressed
Kayak

Write a boolean method that uses recursion to determine whether a `String` argument is a palindrome. The method should return `true` if the argument reads the same forward and backward. Demonstrate the method in a program.

6. Character Counter

Write a method that uses recursion to count the number of times a specific character occurs in an array of characters. Demonstrate the method in a program.



VideoNote

The Recursive
Power Problem

7. Recursive Power Method

Write a method that uses recursion to raise a number to a power. The method should accept two arguments: the number to be raised and the exponent. Assume that the exponent is a nonnegative integer. Demonstrate the method in a program.

8. Sum of Numbers

Write a method that accepts an integer argument and returns the sum of all the integers from 1 up to the number passed as an argument. For example, if 50 is passed as an argument, the method will return the sum of 1, 2, 3, 4, . . . 50. Use recursion to calculate the sum. Demonstrate the method in a program.

9. Ackermann's Function

Ackermann's function is a recursive mathematical algorithm that can be used to test how well a computer performs recursion. Write a method `ackermann(m, n)`, which solves Ackermann's function. Use the following logic in your method:

If $m = 0$ then return $n + 1$
If $n = 0$ then return `ackermann(m - 1, 1)`
Otherwise, return `ackermann(m - 1, ackermann(m, n - 1))`

Test your method in a program that displays the return values of the following method calls:

```
ackermann(0, 0)    ackermann(0, 1)    ackermann(1, 1)    ackermann(1, 2)
ackermann(1, 3)    ackermann(2, 2)    ackermann(3, 2)
```

10. Recursive Population Class

In Programming Challenge 9 of Chapter 4 you wrote a population class that predicts the size of a population of organisms after a number of days. Modify the class so it uses a recursive method instead of a loop to calculate the number of organisms.